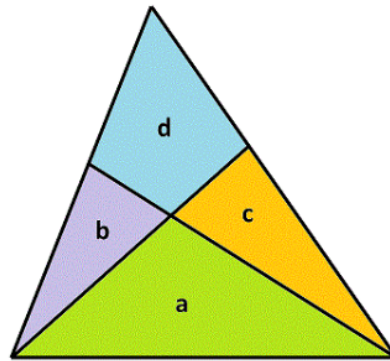


Cutting Triangles

Problem 557



A triangle is cut into four pieces by two straight lines, each starting at one vertex and ending on the opposite edge. This results in forming three smaller triangular pieces, and one quadrilateral. If the original triangle has an integral area, it is often possible to choose cuts such that all of the four pieces also have integral area. For example, the diagram below shows a triangle of area 55 that has been cut in this way.



Representing the areas as a, b, c and d , in the example above, the individual areas are $a = 22, b = 8, c = 11$ and $d = 14$. It is also possible to cut a triangle of area 55 such that $a = 20, b = 2, c = 24, d = 9$.

Define a triangle cutting quadruple (a, b, c, d) as a valid integral division of a triangle, where a is the area of the triangle between the two cut vertices, d is the area of the quadrilateral and b and c are the areas of the two other triangles, with the restriction that $b \leq c$. The two solutions described above are $(22, 8, 11, 14)$ and $(20, 2, 24, 9)$. These are the only two possible quadruples that have a total area of 55.

Define $S(n)$ as the sum of the area of the uncut triangles represented by all valid quadruples with $a + b + c + d \leq n$. For example, $S(20) = 259$.

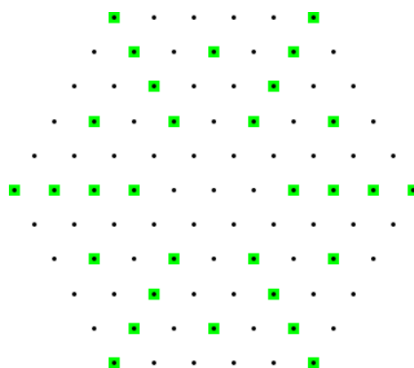
Find $S(10000)$.

Hexagonal Orchards

Problem 351



A *hexagonal orchard* of order n is a triangular lattice made up of points within a regular hexagon with side n . The following is an example of a hexagonal orchard of order 5:



Highlighted in green are the points which are hidden from the center by a point closer to it. It can be seen that for a hexagonal orchard of order 5, 30 points are hidden from the center.

Let $H(n)$ be the number of points hidden from the center in a hexagonal orchard of order n .

$H(5) = 30, H(10) = 138, H(1000) = 1177848$.

Find $H(100\,000\,000)$.

Beautiful Graphs

Problem 857



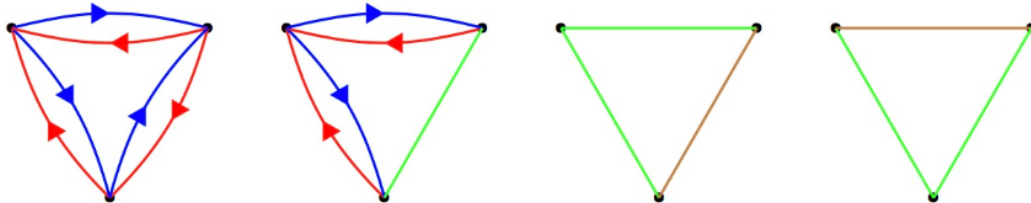
A graph is made up of vertices and coloured edges. Between every two distinct vertices there must be exactly one of the following:

- A red directed edge one way, and a blue directed edge the other way
- A green undirected edge
- A brown undirected edge

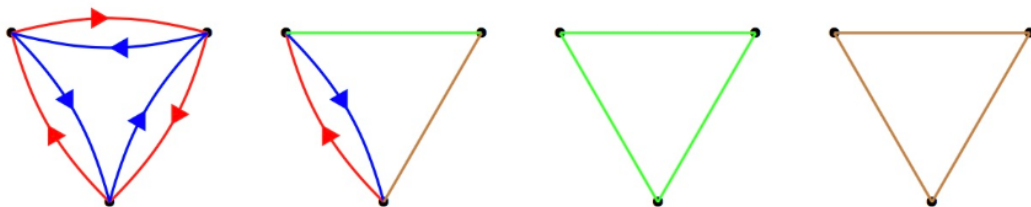
Such a graph is called *beautiful* if

- A cycle of edges contains a red edge **if and only if** it also contains a blue edge
- No triangle of edges is made up of entirely green or entirely brown edges

Below are four distinct examples of beautiful graphs on three vertices:



Below are four examples of graphs that are not beautiful:



Let $G(n)$ be the number of beautiful graphs on the labelled vertices: $1, 2, \dots, n$. You are given $G(3) = 24$, $G(4) = 186$ and $G(15) = 12472315010483328$.

Find $G(10^7)$. Give your answer modulo $10^9 + 7$.

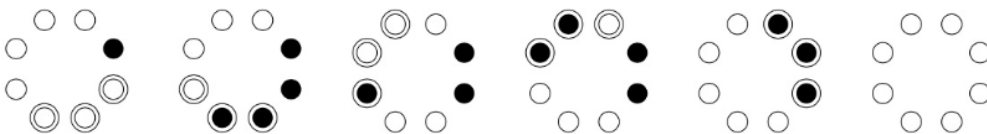
Circle of Coins

Problem 728



Consider n coins arranged in a circle where each coin shows heads or tails. A move consists of turning over k consecutive coins: tail-head or head-tail. Using a sequence of these moves the objective is to get all the coins showing heads.

Consider the example, shown below, where $n = 8$ and $k = 3$ and the initial state is one coin showing tails (black). The example shows a solution for this state.



For given values of n and k not all states are solvable. Let $F(n, k)$ be the number of states that are solvable. You are given that $F(3, 2) = 4$, $F(8, 3) = 256$ and $F(9, 3) = 128$.

Further define:

$$S(N) = \sum_{n=1}^N \sum_{k=1}^n F(n, k).$$

You are also given that $S(3) = 22$, $S(10) = 10444$ and $S(10^3) \equiv 853837042 \pmod{1\,000\,000\,007}$.

Find $S(10^7)$. Give your answer modulo $1\,000\,000\,007$.

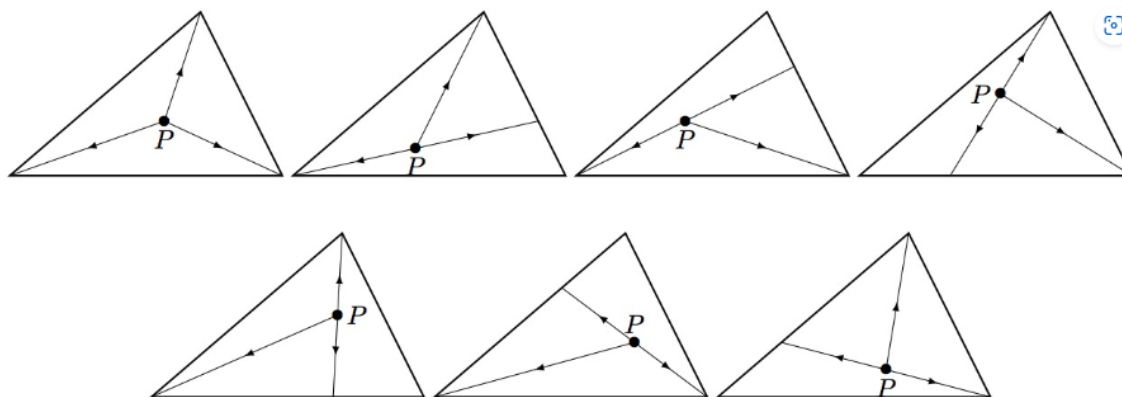
Triangular Pizza

Problem 747



Mamma Triangolo baked a triangular pizza. She wants to cut the pizza into n pieces. She first chooses a point P in the interior (not boundary) of the triangle pizza, and then performs n cuts, which all start from P and extend straight to the boundary of the pizza so that the n pieces are all triangles and all have the same area.

Let $\psi(n)$ be the number of different ways for Mamma Triangolo to cut the pizza, subject to the constraints. For example, $\psi(3) = 7$.



Also $\psi(6) = 34$, and $\psi(10) = 90$.

Let $\Psi(m) = \sum_{n=3}^m \psi(n)$. You are given $\Psi(10) = 345$ and $\Psi(1000) = 172166601$.

Find $\Psi(10^8)$. Give your answer modulo 1 000 000 007.

Cutting Rectangular Grid Paper

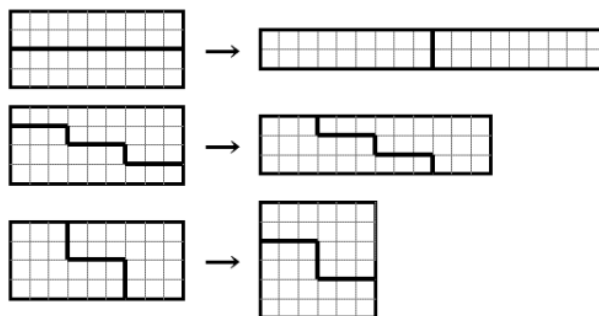
Problem 338



A rectangular sheet of grid paper with integer dimensions $w \times h$ is given. Its grid spacing is 1.

When we cut the sheet along the grid lines into two pieces and rearrange those pieces without overlap, we can make new rectangles with different dimensions.

For example, from a sheet with dimensions 9×4 , we can make rectangles with dimensions 18×2 , 12×3 and 6×6 by cutting and rearranging as below:



Similarly, from a sheet with dimensions 9×8 , we can make rectangles with dimensions 18×4 and 12×6 .

For a pair w and h , let $F(w, h)$ be the number of distinct rectangles that can be made from a sheet with dimensions $w \times h$.

For example, $F(2, 1) = 0$, $F(2, 2) = 1$, $F(9, 4) = 3$ and $F(9, 8) = 2$.

Note that rectangles congruent to the initial one are not counted in $F(w, h)$.

Note also that rectangles with dimensions $w \times h$ and dimensions $h \times w$ are not considered distinct.

For an integer N , let $G(N)$ be the sum of $F(w, h)$ for all pairs w and h which satisfy $0 < h \leq w \leq N$.

We can verify that $G(10) = 55$, $G(10^3) = 971745$ and $G(10^5) = 9992617687$.

Find $G(10^{12})$. Give your answer modulo 10^8 .

Protein Folding

Problem 300



In a very simplified form, we can consider proteins as strings consisting of hydrophobic (H) and polar (P) elements, e.g. HHPPHHHPHHPH.

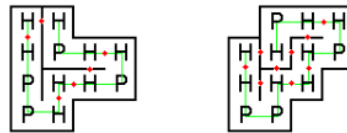
For this problem, the orientation of a protein is important; e.g. HPP is considered distinct from PPH. Thus, there are 2^n distinct proteins consisting of n elements.

When one encounters these strings in nature, they are always folded in such a way that the number of H-H contact points is as large as possible, since this is energetically advantageous.

As a result, the H-elements tend to accumulate in the inner part, with the P-elements on the outside.

Natural proteins are folded in three dimensions of course, but we will only consider protein folding in two dimensions.

The figure below shows two possible ways that our example protein could be folded (H-H contact points are shown with red dots).



The folding on the left has only six H-H contact points, thus it would never occur naturally.

On the other hand, the folding on the right has nine H-H contact points, which is optimal for this string.

Assuming that H and P elements are equally likely to occur in any position along the string, the average number of H-H contact points in an optimal folding of a random protein string of length 8 turns out to be $850/2^8 = 3.3203125$.

What is the average number of H-H contact points in an optimal folding of a random protein string of length 15?

Give your answer using as many decimal places as necessary for an exact result.